

THEOREM 1—Fubini's Theorem (First Form) If $f(x, y)$ is continuous throughout the rectangular region $R: a \leq x \leq b, c \leq y \leq d$, then

$$\iint_R f(x, y) dA = \int_c^d \int_a^b f(x, y) dx dy = \int_a^b \int_c^d f(x, y) dy dx.$$

Fubini's Theorem says that double integrals over rectangles can be calculated as iterated integrals. Thus, we can evaluate a double integral by integrating with respect to one variable at a time.

Fubini's Theorem also says that we may calculate the double integral by integrating in *either* order, a genuine convenience. When we calculate a volume by slicing, we may use either planes perpendicular to the x -axis or planes perpendicular to the y -axis.

EXAMPLE 1 Calculate $\iint_R f(x, y) dA$ for

$$f(x, y) = 100 - 6x^2y \quad \text{and} \quad R: 0 \leq x \leq 2, -1 \leq y \leq 1.$$

Solution Figure 15.6 displays the volume beneath the surface. By Fubini's Theorem,

$$\begin{aligned} \iint_R f(x, y) dA &= \int_{-1}^1 \int_0^2 (100 - 6x^2y) dx dy = \int_{-1}^1 [100x - 2x^3y]_{x=0}^{x=2} dy \\ &= \int_{-1}^1 (200 - 16y) dy = [200y - 8y^2]_{-1}^1 = 400. \end{aligned}$$

Reversing the order of integration gives the same answer:

$$\begin{aligned} \int_0^2 \int_{-1}^1 (100 - 6x^2y) dy dx &= \int_0^2 [100y - 3x^2y^2]_{y=-1}^{y=1} dx \\ &= \int_0^2 [(100 - 3x^2) - (-100 - 3x^2)] dx \\ &= \int_0^2 200 dx = 400. \end{aligned}$$

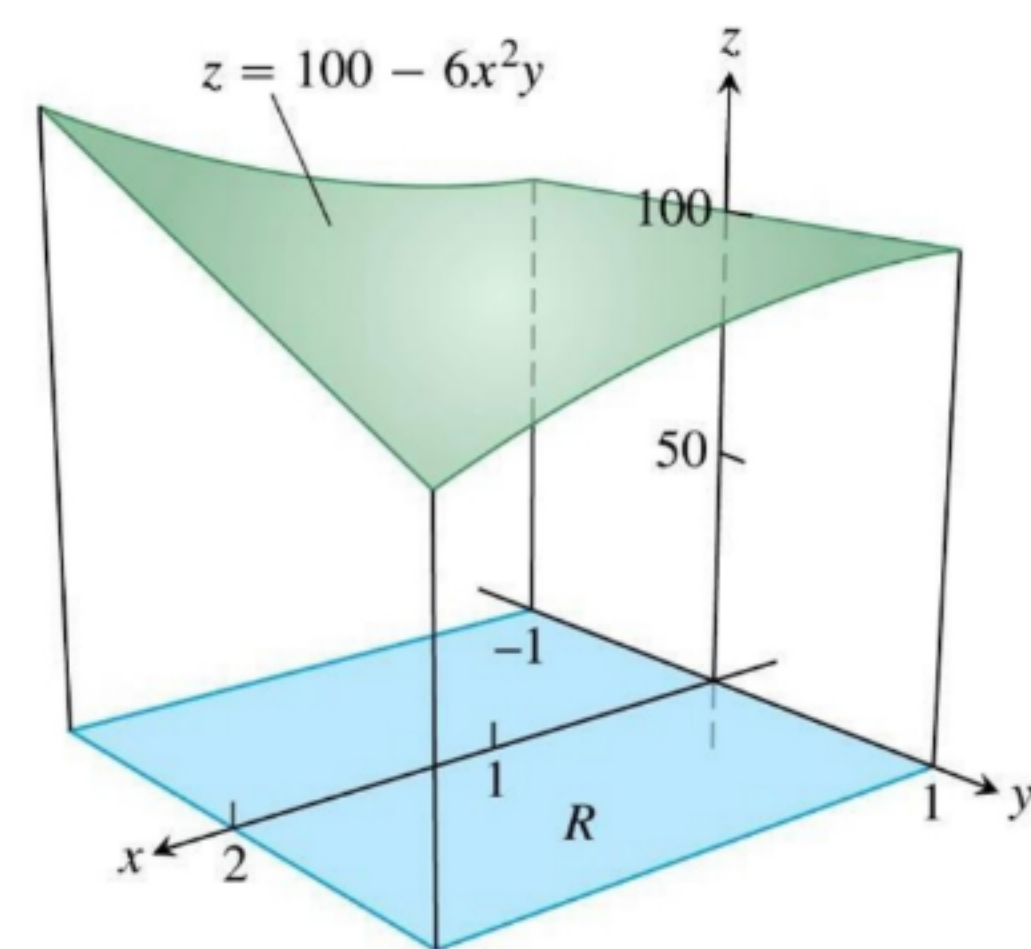


FIGURE 15.6 The double integral $\iint_R f(x, y) dA$ gives the volume under this surface over the rectangular region R (Example 1).

THEOREM 2—Fubini's Theorem (Stronger Form) Let $f(x, y)$ be continuous on a region R .

1. If R is defined by $a \leq x \leq b, g_1(x) \leq y \leq g_2(x)$, with g_1 and g_2 continuous on $[a, b]$, then

$$\iint_R f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx.$$

2. If R is defined by $c \leq y \leq d, h_1(y) \leq x \leq h_2(y)$, with h_1 and h_2 continuous on $[c, d]$, then

$$\iint_R f(x, y) dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy.$$

EXAMPLE 1 Find the volume of the prism whose base is the triangle in the xy -plane bounded by the x -axis and the lines $y = x$ and $x = 1$ and whose top lies in the plane

$$z = f(x, y) = 3 - x - y.$$

Solution See Figure 15.12. For any x between 0 and 1, y may vary from $y = 0$ to $y = x$ (Figure 15.12b). Hence,

$$\begin{aligned} V &= \int_0^1 \int_0^x (3 - x - y) dy dx = \int_0^1 \left[3y - xy - \frac{y^2}{2} \right]_{y=0}^{y=x} dx \\ &= \int_0^1 \left(3x - \frac{3x^2}{2} \right) dx = \left[\frac{3x^2}{2} - \frac{x^3}{2} \right]_{x=0}^{x=1} = 1. \end{aligned}$$

When the order of integration is reversed (Figure 15.12c), the integral for the volume is

$$\begin{aligned} V &= \int_0^1 \int_y^1 (3 - x - y) dx dy = \int_0^1 \left[3x - \frac{x^2}{2} - xy \right]_{x=y}^{x=1} dy \\ &= \int_0^1 \left(3 - \frac{1}{2} - y - 3y + \frac{y^2}{2} + y^2 \right) dy \\ &= \int_0^1 \left(\frac{5}{2} - 4y + \frac{3}{2}y^2 \right) dy = \left[\frac{5}{2}y - 2y^2 + \frac{y^3}{2} \right]_{y=0}^{y=1} = 1. \end{aligned}$$

The two integrals are equal, as they should be.

Properties of Double Integrals

If $f(x, y)$ and $g(x, y)$ are continuous on the bounded region R , then the following properties hold.

1. *Constant Multiple:* $\iint_R cf(x, y) dA = c \iint_R f(x, y) dA$ (any number c)

2. *Sum and Difference:*

$$\iint_R (f(x, y) \pm g(x, y)) dA = \iint_R f(x, y) dA \pm \iint_R g(x, y) dA$$

3. *Domination:*

(a) $\iint_R f(x, y) dA \geq 0$ if $f(x, y) \geq 0$ on R

(b) $\iint_R f(x, y) dA \geq \iint_R g(x, y) dA$ if $f(x, y) \geq g(x, y)$ on R

4. *Additivity:* $\iint_R f(x, y) dA = \iint_{R_1} f(x, y) dA + \iint_{R_2} f(x, y) dA$

if R is the union of two nonoverlapping regions R_1 and R_2

EXAMPLE 4 Find the volume of the wedgelike solid that lies beneath the surface $z = 16 - x^2 - y^2$ and above the region R bounded by the curve $y = 2\sqrt{x}$, the line $y = 4x - 2$, and the x -axis.

Solution Figure 15.18a shows the surface and the “wedgelike” solid whose volume we want to calculate. Figure 15.18b shows the region of integration in the xy -plane. If we integrate in the order $dy dx$ (first with respect to y and then with respect to x), two integrations will be required because y varies from $y = 0$ to $y = 2\sqrt{x}$ for $0 \leq x \leq 0.5$, and then varies from $y = 4x - 2$ to $y = 2\sqrt{x}$ for $0.5 \leq x \leq 1$. So we choose to integrate in the order $dx dy$, which requires only one double integral whose limits of integration are indicated in Figure 15.18b. The volume is then calculated as the iterated integral:

$$\begin{aligned} \iint_R (16 - x^2 - y^2) dA &= \int_0^2 \int_{y^2/4}^{(y+2)/4} (16 - x^2 - y^2) dx dy \\ &= \int_0^2 \left[16x - \frac{x^3}{3} - xy^2 \right]_{x=y^2/4}^{x=(y+2)/4} dy \\ &= \int_0^2 \left[4(y+2) - \frac{(y+2)^3}{3 \cdot 64} - \frac{(y+2)y^2}{4} - 4y^2 + \frac{y^6}{3 \cdot 64} + \frac{y^4}{4} \right] dy \\ &= \left[\frac{191y}{24} + \frac{63y^2}{32} - \frac{145y^3}{96} - \frac{49y^4}{768} + \frac{y^5}{20} + \frac{y^7}{1344} \right]_0^2 = \frac{20803}{1680} \approx 12.4. \end{aligned}$$

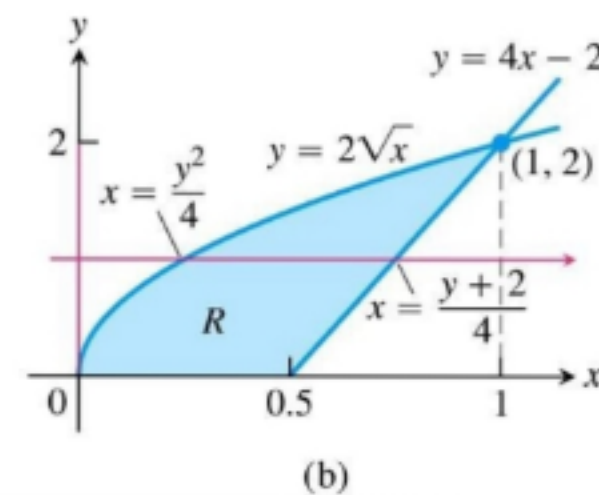
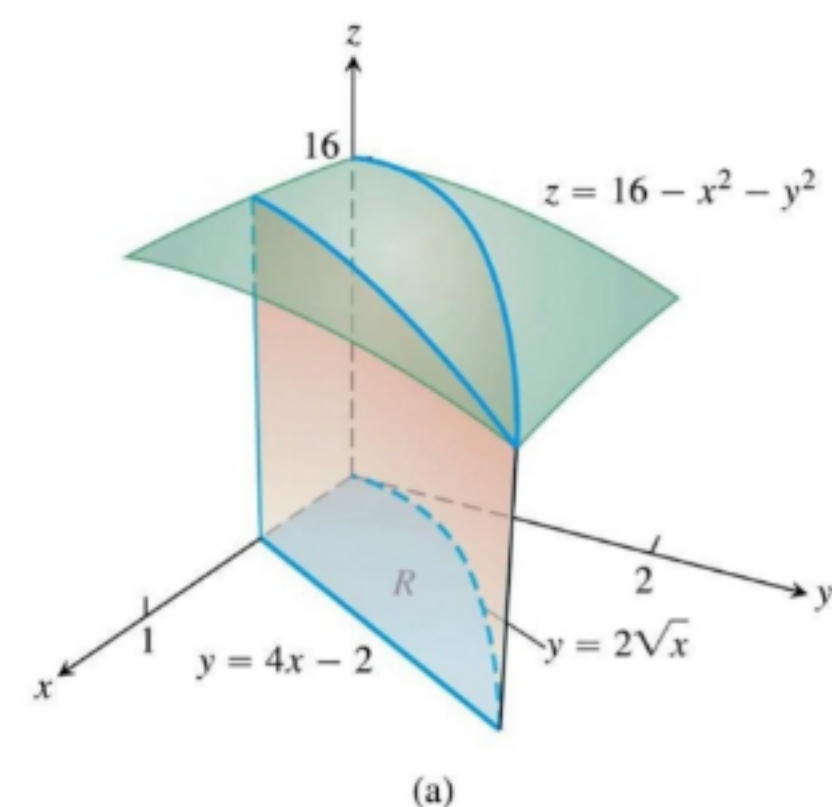


FIGURE 15.18 (a) The solid “wedgelike” region whose volume is found in Example 4. (b) The region of integration R showing the order $dx dy$.

Areas of Bounded Regions in the Plane

If we take $f(x, y) = 1$ in the definition of the double integral over a region R in the preceding section, the Riemann sums reduce to

$$S_n = \sum_{k=1}^n f(x_k, y_k) \Delta A_k = \sum_{k=1}^n \Delta A_k. \quad (1)$$

This is simply the sum of the areas of the small rectangles in the partition of R , and approximates what we would like to call the area of R . As the norm of a partition of R approaches zero, the height and width of all rectangles in the partition approach zero, and the coverage of R becomes increasingly complete (Figure 15.8). We define the area of R to be the limit

$$\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n \Delta A_k = \iint_R dA. \quad (2)$$

DEFINITION The **area** of a closed, bounded plane region R is

$$A = \iint_R dA.$$

EXAMPLE 1 Find the area of the region R bounded by $y = x$ and $y = x^2$ in the first quadrant.

Solution We sketch the region (Figure 15.19), noting where the two curves intersect at the origin and $(1, 1)$, and calculate the area as

$$\begin{aligned} A &= \int_0^1 \int_{x^2}^x dy dx = \int_0^1 \left[y \right]_{x^2}^x dx \\ &= \int_0^1 (x - x^2) dx = \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = \frac{1}{6}. \end{aligned}$$

Notice that the single-variable integral $\int_0^1 (x - x^2) dx$, obtained from evaluating the inside iterated integral, is the integral for the area between these two curves using the method of Section 5.6.

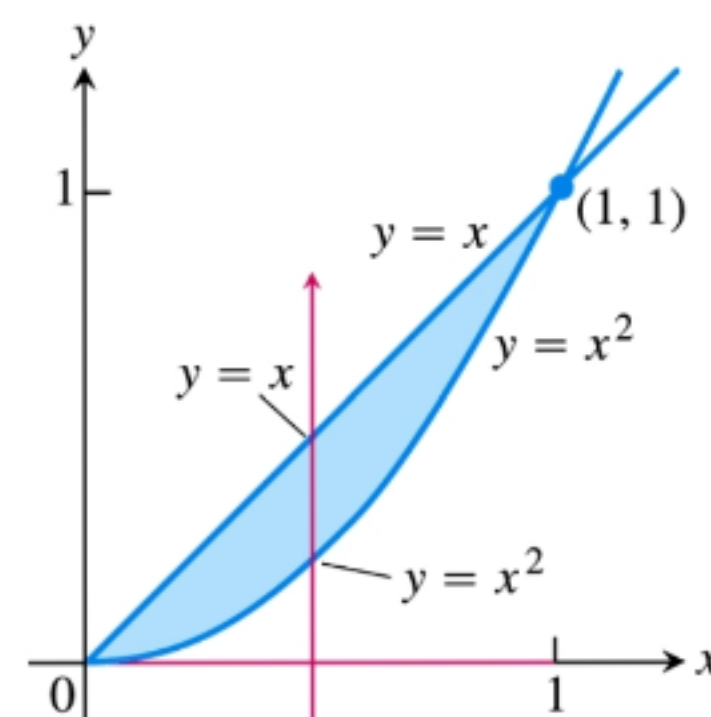


FIGURE 15.19 The region in Example 1.

EXAMPLE 2 Find the area of the region R enclosed by the parabola $y = x^2$ and the line $y = x + 2$.

Solution If we divide R into the regions R_1 and R_2 shown in Figure 15.20a, we may calculate the area as

$$A = \iint_{R_1} dA + \iint_{R_2} dA = \int_0^1 \int_{-\sqrt{y}}^{\sqrt{y}} dx \, dy + \int_1^4 \int_{y-2}^{\sqrt{y}} dx \, dy.$$

On the other hand, reversing the order of integration (Figure 15.20b) gives

$$A = \int_{-1}^2 \int_{x^2}^{x+2} dy \, dx.$$

This second result, which requires only one integral, is simpler and is the only one we would bother to write down in practice. The area is

$$A = \int_{-1}^2 \left[y \right]_{x^2}^{x+2} dx = \int_{-1}^2 (x + 2 - x^2) dx = \left[\frac{x^2}{2} + 2x - \frac{x^3}{3} \right]_{-1}^2 = \frac{9}{2}.$$

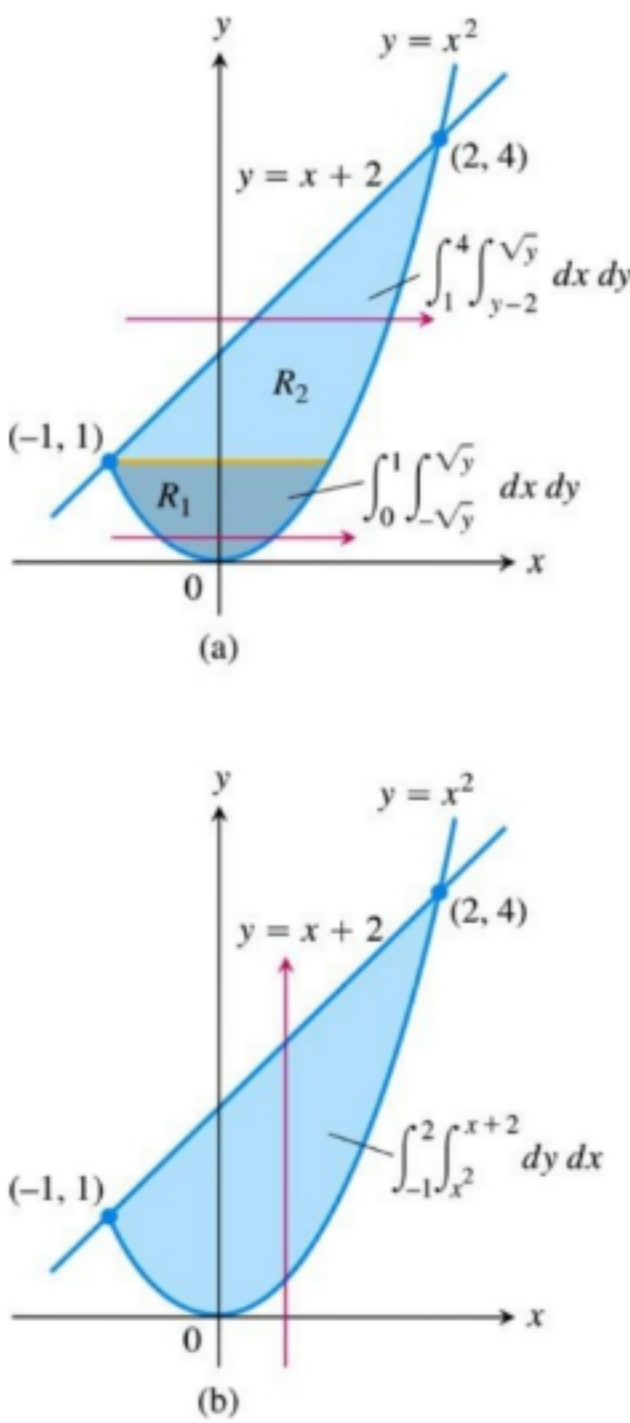


FIGURE 15.20 Calculating this area takes (a) two double integrals if the first integration is with respect to x , but (b) only one if the first integration is with respect to y (Example 2).

Average value of f over R
 $= \frac{1}{\text{area of } R} \iint_R f \, dA.$

(3)

EXAMPLE 3 Find the average value of $f(x, y) = x \cos xy$ over the rectangle R : $0 \leq x \leq \pi$, $0 \leq y \leq 1$.

Solution The value of the integral of f over R is

$$\begin{aligned} \int_0^\pi \int_0^1 x \cos xy \, dy \, dx &= \int_0^\pi \left[\sin xy \right]_{y=0}^{y=1} dx & \int x \cos xy \, dy &= \sin xy + C \\ &= \int_0^\pi (\sin x - 0) \, dx = -\cos x \Big|_0^\pi = 1 + 1 = 2. \end{aligned}$$

The area of R is π . The average value of f over R is $2/\pi$.

Evaluating Iterated Integrals

In Exercises 1–12, evaluate the iterated integral.

1. $\int_1^2 \int_0^4 2xy \, dy \, dx$

2. $\int_0^2 \int_{-1}^1 (x - y) \, dy \, dx$

Evaluating Double Integrals over Rectangles

In Exercises 13–20, evaluate the double integral over the given region R .

13. $\iint_R (6y^2 - 2x) \, dA, \quad R: 0 \leq x \leq 1, \quad 0 \leq y \leq 2$

14. $\iint_R \left(\frac{\sqrt{x}}{y^2} \right) \, dA, \quad R: 0 \leq x \leq 4, \quad 1 \leq y \leq 2$

15. $\iint_R xy \cos y \, dA, \quad R: -1 \leq x \leq 1, \quad 0 \leq y \leq \pi$

Volume Beneath a Surface

23. Find the volume of the region bounded above by the paraboloid $z = x^2 + y^2$ and below by the square $R: -1 \leq x \leq 1, -1 \leq y \leq 1$.

24. Find the volume of the region bounded above by the elliptical paraboloid $z = 16 - x^2 - y^2$ and below by the square $R: 0 \leq x \leq 2, 0 \leq y \leq 2$.

25. Find the volume of the region bounded above by the plane $z = 2 - x - y$ and below by the square $R: 0 \leq x \leq 1, 0 \leq y \leq 1$.

In Exercises 25–28, integrate f over the given region.

25. **Quadrilateral** $f(x, y) = x/y$ over the region in the first quadrant bounded by the lines $y = x, y = 2x, x = 1$, and $x = 2$

26. **Triangle** $f(x, y) = x^2 + y^2$ over the triangular region with vertices $(0, 0), (1, 0)$, and $(0, 1)$